RATISS Verse I

A sealed, falsifiable reference atlas of ten cosmological universes

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Abstract

We present Ratiss Verse I, a reference atlas in which ten distinct cosmological models are integrated through a *single* solver chain: one background quadrature, one linear growth equation, and one particle–mesh N-body engine. Every number in the atlas is recomputed from a frozen parameter set at build time; nothing is transcribed by hand; every produced artefact is sealed with a SHA-256 manifest.

The atlas is built to answer one question honestly: *which of these ten universes is actually distinguishable from ΛCDM?* Our answer is uncomfortable. Six models — ΛCDM, $f(R)$ Hu–Sawicki, $f(T)$ teleparallel, a running-vacuum model with $\nu = 10^{-3}$, a constant-w fluid at $w = -1.004$, and the RATISS granular universe — agree on $H_0 t_0$ to within 5.6×10^{-4} (0.06%). The dimensionless age is therefore *degenerate* and cannot serve as a discriminator. The growth index γ separates them at the 6.7% level ($0.5542 \to 0.5916$), which is within reach of stage-IV surveys.

We report four analytic results and one negative result of some weight. The negative result concerns the widely quoted target $H_0 t_0 = 0.9549$: we show it is not a property of any equation of state but of *emptiness*. A universe containing a single substance with $\rho \propto a^{-n}$ has $H_0 t_0 = 2/n$ exactly; with $n = 2.0945$ this gives 0.95488. Adding the matter that galaxies require drives $H_0 t_0$ down to 0.7927, an age of 11.51 Gyr, below the age of the oldest stars. Reproducing 0.9549 with matter present requires a *phantom* equation of state, $w = -1.0332$, or $\Omega_{\rm m} = 0.3100$.

We also report the defects we found in our own pipeline, including a radiation-inconsistency between model families that shifted $H_0 t_0$ by 3.8×10^{-4} — larger than several effects the atlas was built to resolve — and we give the corrected values throughout.

Every figure, table and hash in this document can be regenerated with one command. The repository is the artefact.

Keywords: cosmology — dark energy — modified gravity — N-body simulation — reproducibility — falsifiability — open science

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1 Introduction

1.1 The problem this atlas addresses

Cosmological model building has an asymmetry problem. It is cheap to write down a modification of general relativity that fits the expansion history; it is expensive, and rarely done, to state in advance what measurement would kill it. The literature contains a large number of models whose predictions differ from ΛCDM at a level the models themselves do not quantify.

This atlas was built to make that quantification mechanical. Ten models are declared. Each is reduced to three ingredients:

(i) a background expansion rate $E(a) = H(a)/H_0$;
(ii) a linear growth history, driven by an effective coupling $\mu(k, a)$;
(iii) a local force law, entering the N-body engine through the same $\mu(k, a)$.

Given those three, every observable quantity used here follows from two solvers and one simulation. No model is allowed a bespoke derivation. This is the central design decision: *one code path, ten universes.*

1.2 Lineage

The atlas is the direct successor of a toy model, Syntoron 24, which integrated 24 particles under the pair force

$$F(r) = -\frac{G}{r^2} + \frac{K}{r^4}, \qquad r_c \equiv \sqrt{K/G}, \tag{1}$$

and reported a single dimensionless number, $Ht = 1$. The toy was internally consistent and externally unfalsifiable: it had no expansion history, no spectrum, and no observational contact.

Ratiss Verse I keeps the force law of Eq. (1) — it is the only content of the toy that survived — and embeds it in a cosmological setting where it has consequences (Section 6.3). Everything else in the toy is retired.

1.3 What is new here

- A *registry* of ten universes sharing one solver chain, with every number regenerated at build time (`scripts/build.py`, 26.6 s wall clock).
- Three analytic results (Section 4) that constrain what a cut-off can and cannot do.
- A negative result of general interest: the 0.9549 target that motivated a family of single-fluid models is an artefact of the absence of matter (Section 8).
- A documented list of the defects found in our own pipeline, with the corrected numbers (Section 9 and Appendix B).
- Cryptographic sealing: twelve artefacts, SHA-256 manifest, verification in one command, twenty unit tests (Section 10).

1.4 Conventions

Unless stated otherwise we use Planck 2018 TT,TE,EE+lowE+lensing values [1]: $\Omega_{\rm m} = 0.3153$, $\Omega_\Lambda = 1 - \Omega_{\rm m} - \Omega_{\rm r}$, $\Omega_{\rm r} = 9.15 \times 10^{-5}$ (photons plus three massless neutrinos), $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$ ($h = 0.6736$), $\sigma_8 = 0.8111$, $n_s = 0.9649$. Time is in units of H_0^{-1}; the present epoch is $a = 1$. The dimensionless age is $H_0 t_0 = \int_0^1 {\rm d}a\,[aE(a)]^{-1}$. Radiation is retained in *every* background of the registry (Section 9 explains why this sentence had to be added).

2 The framework

Definition 2.1 (Reference universe)**.** A reference universe is a triple $\mathcal{U} = \langle E, \mu, \mathcal{L} \rangle$ where $E(a)$ is the expansion rate, $\mu(k, a)$ the effective gravitational coupling of linear perturbations relative to Newton, and $\mathcal{L}$ a local force law.

Two universes are *operationally identical in this atlas* if they share all three. Two universes are *degenerate* here if they produce the same value of a given observable, regardless of whether they differ physically. The distinction matters: Section 6 contains universes that are degenerate by measurement and universes that are degenerate by construction, and we label them separately.

2.1 Background

All quantities are derived from $E(a)$ by quadrature:

$$H_0 t_0 = \int_0^1 \frac{\mathrm{d}a}{a\,E(a)}, \tag{2}$$

$$t_0 = \frac{H_0 t_0}{H_0}, \tag{3}$$

$$D_C(z) = \frac{c}{H_0} \int_0^z \frac{\mathrm{d}z'}{E(z')}. \tag{4}$$

Equation (2) is the quantity that motivated this project, and Section 8 is devoted to it.

We integrate in $\ln a$ from $a = e^{-60}$ to $a = 1$ using adaptive quadrature with two tolerance settings; the reported uncertainty is the difference between the two runs plus the first run's own error estimate. It is a *quadrature* error of order 10^{-14}, not a physical uncertainty. Conflating the two is one of the defects catalogued in Section 9.

2.2 Linear growth

The growth of matter perturbations obeys, with $N = \ln a$ and $' \equiv \mathrm{d}/\mathrm{d}N$,

$$D'' + \left(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\right) D' - \frac{3}{2}\,\Omega_{\mathrm{m}}(N)\,\mu(k,a)\,D = 0, \tag{5}$$

with $\Omega_{\mathrm{m}}(N) = \Omega_{\mathrm{m}}\,a^{-3}/E^2(a)$ and $\mu = 1$ in general relativity. The growth index is

$$\gamma \equiv \frac{\ln f}{\ln \Omega_{\mathrm{m}}}, \qquad f \equiv \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a}. \tag{6}$$

Equation (5) is solved with LSODA at $\mathrm{rtol} = 10^{-11}$, $\mathrm{atol} = 10^{-15}$, from $N_{\mathrm{init}} = -9$ with the pure matter-era growing mode $D = D' = a$ as initial condition. That choice is an approximation and is listed as defect D5 in Appendix B: it discards any radiation-era and any early modified-gravity transient.

The predicted observable is $f\sigma_8(z) = f(z)\,D(z)/D(0)\,\sigma_8$, with σ_8 held fixed at the Planck value. Holding it fixed means the atlas predicts the *shape* of $f\sigma_8(z)$, not its normalisation: an overall amplitude rescaling is degenerate with σ_8 and this atlas does not claim otherwise.

2.3 The granular coupling

The force law of Eq. (1) factorises exactly as

$$F(r) = -\frac{G}{r^2}\left[1 - \left(\frac{r_c}{r}\right)^2\right], \tag{7}$$

Table 1: The ten universes of the registry. $E(a) = H(a)/H_0$; $\Omega_{rc} = [(1-\Omega_{\rm m})/2]^2$; ν is the running-vacuum coefficient; m is the Cardassian index. All values of $\Omega_{\rm m}$ are the Planck value except where stated.

Key	Family	$E^2(a)$ or $E(a)$	Reference
`lcdm`	GR	$\Omega_{\rm m}a^{-3} + \Omega_{\rm r}a^{-4} + \Omega_\Lambda$	[1]
`w_m0302`	fluid	$\Omega_{\rm m}a^{-3} + \Omega_{\rm r}a^{-4} + \Omega_x a^{-3(1+w)}$, $w = -0.30185$	—
`w_m1004`	fluid	same, $w = -1.004049$	—
`rvm_001`	running vacuum	$[\Omega_{\rm m}a^{-3} + 1 - \Omega_{\rm m} - \nu]/(1-\nu)$, $\nu = 10^{-3}$	[8]
`dgp`	braneworld	$\left[\Omega_{\rm m}a^{-3} + \Omega_{\rm r}a^{-4} + \Omega_{rc}\right]^{1/2} + \Omega_{rc}^{1/2}$	[4]
`cardassian_05`	mod. Friedmann	$\Omega_{\rm m}a^{-3} + \Omega_{\rm r}a^{-4} + \Omega_\Lambda a^{-m}$, $m = 0.5$	[7]
`fr_hs`	$f(R)$	= `lcdm` (designer)	[5]
`ft`	$f(T)$	= `lcdm` (designer)	[6]
`granular`	RATISS	= `lcdm` (designer)	this work
`fluid_n2094`	single fluid	$a^{-n/2}$, $n = 2.0945$, nothing else	—

so that $r \gg r_c$ is Newtonian, $r = r_c$ is force-free, and $r < r_c$ is repulsive. The corresponding relative force deficit is $1 - \mu = (r_c/r)^2$, a two-derivative correction.

The exact Fourier transform of a $1/r^4$ force is not a tempered distribution: it diverges logarithmically in the infrared. This is not a technicality — it is the reason the model *requires* a cut-off and cannot exist without one. The minimal bounded regularisation that reproduces the three exact limits is

$$\mu(x) = \frac{1 - x^2}{1 + x^2}, \qquad x \equiv k\, r_c, \tag{8}$$

with $\mu(0^+) = +1$, $\mu(1) = 0$, $\mu(\infty) = -1$. We state plainly that Eq. (8) is an *interpolation*, not a derived transform, and that the mapping from r_c to x is a modelling assumption (defect D7).

3 The registry

Table 1 lists the ten universes with their Friedmann equations. The classification column is computed, not assigned by hand: a model is marked EXCLUDED when $|H_0 t_0 - H_0 t_0^{\Lambda\rm CDM}| > 0.02$, TESTABLE when $|\gamma - \gamma^{\Lambda\rm CDM}| > 0.03$, and DEGENERATE otherwise. Those two thresholds are chosen (Section 9), and the reader should treat the labels as ordering devices rather than verdicts.

Three of these rows deserve a warning that the table cannot express.

`fr_hs` and `ft` are degenerate by construction. Both are assigned $E = E_{\Lambda\rm CDM}$ *and* $\mu = 1$ throughout. Their growth index is therefore, identically, the ΛCDM value. Labelling them DEGENERATE is vacuous: they were never given the chance to differ. A genuine $f(R)$ computation requires a scale- and environment-dependent $\mu(k, a)$ with chameleon screening, which this atlas does not implement. The rows are retained because the registry is meant to be extended, not because they constitute evidence of anything.

`granular` is also degenerate by construction in the registry. The registry row is evaluated at $\mu = 1$; the cut-off appears only in the dedicated sector of Section 6.3. The row exists so that the background column is complete, and its DEGENERATE label carries no information about the granular model.

`fluid_n2094` has no matter in it. It is a single substance and nothing else. It is included precisely because it is the cleanest possible illustration of Section 8.

Table 2: Predictions of the ten universes, recomputed at build time. $\Delta H_0 t_0$ is relative to the ΛCDM reference. $f\sigma_8(z{=}1)$ uses $\sigma_8 = 0.8111$ today.

Universe	$H_0 t_0$	$\Delta H_0 t_0$	γ	$f\sigma_8(1)$	Status
`lcdm`	0.950351	0	0.5544	0.4314	degenerate
`w_m0302`	0.792674	-1.577×10^{-1}	0.5916	0.3478	excluded
`w_m1004`	0.950915	$+5.64 \times 10^{-4}$	0.5542	0.4316	degenerate
`rvm_001`	0.950462	$+1.11 \times 10^{-4}$	0.5544	0.4313	degenerate
`dgp`	0.894863	-5.55×10^{-2}	0.5663	0.4021	excluded
`cardassian_05`	0.924667	-2.57×10^{-2}	0.5602	0.4196	excluded
`fr_hs`	0.950351	0	0.5544	0.4314	degenerate†
`ft`	0.950351	0	0.5544	0.4314	degenerate†
`granular`	0.950351	0	0.5544	0.4314	alive–but–invisible†
`fluid_n2094`	0.954882	$+4.53 \times 10^{-3}$	0.5706	0.3962	excluded

†Degenerate by construction, not by measurement (see text).

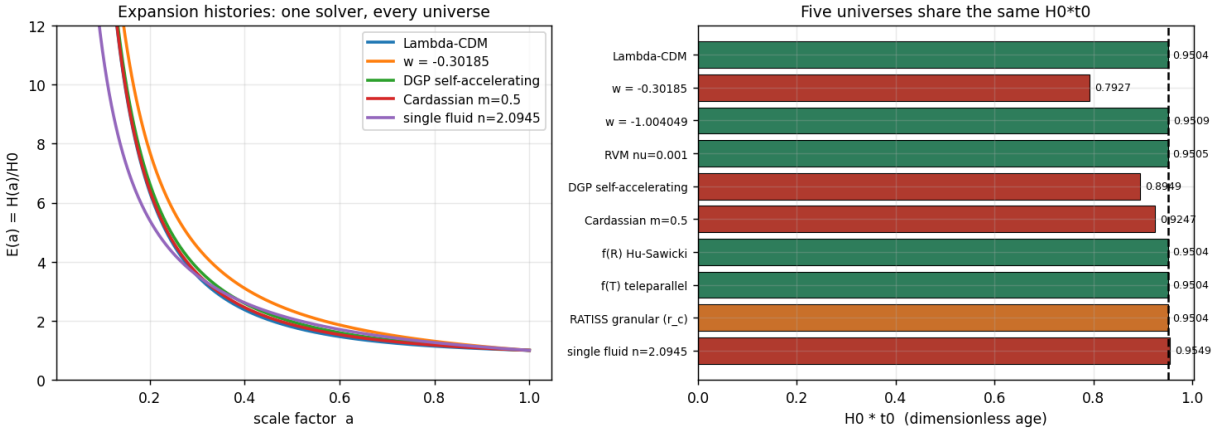


Figure 1: Left: expansion histories of all ten universes, one solver. Right: the universes that share the same dimensionless age. The vertical spread on the left panel is large; the spread in $H_0 t_0$ among the six degenerate models is 5.6×10^{-4}.

3.1 Results from the registry

Table 2 gives the computed values. The reference is

$$H_0 t_0^{\Lambda\mathrm{CDM}} = 0.950350929653681, \tag{9}$$

which converts, with $H_0^{-1} = 14.5159$ Gyr, to

$$t_0 = 13.7952 \text{ Gyr}. \tag{10}$$

Planck 2018 reports 13.801 ± 0.024 Gyr [1]. The agreement, $\Delta = -0.006$ Gyr, is an *independent* check of the whole background pipeline: this number was produced by quadrature over $E(a)$, not copied. We regard it as the single most important validation in this work, because it is a comparison against a number we did not compute.

4 Analytic results

Four results are proved, not measured. They are what makes the numerical tables interpretable.

Table 3: The level set $H_0 t_0 = 0.9549$ in two different one-parameter families. Both entries were obtained by root finding on the *corrected* backgrounds (radiation included).

Family	Parameter that gives $H_0 t_0 = 0.9549$	Comment
matter + fluid, $\Omega_{\rm m} = 0.3153$	$w = -1.0332$	phantom
ΛCDM, $w = -1$	$\Omega_{\rm m} = 0.3100$	viable
single fluid, no matter	$n = 2.0945$	no galaxies

4.1 The mean density exerts no force

Theorem 4.1 (Jeans swindle, stated as a simulation constraint)**.** *In a periodic box of side L with mean density $\bar\rho$, the Poisson equation $\nabla^2\Phi = 4\pi G(\rho - \bar\rho)$ has no $\mathbf{k} = 0$ mode. The uniform component of the density therefore produces no acceleration, and the expansion rate $a(t)$ cannot be generated by the simulated particles.*

Proof. Fourier transforming, $-k^2\tilde\Phi_{\mathbf{k}} = 4\pi G\,\tilde\delta_{\mathbf{k}}$. For $\mathbf{k} = 0$ the left-hand side vanishes identically while $\tilde\delta_0 = 0$ by construction of the overdensity. No solution with $\tilde\Phi_0 \neq 0$ exists. □

Remark 4.2. *This is the reason every production N-body code supplies $a(t)$ from a Friedmann solver. In this atlas the solver is* `background.py`*; the particles integrate only the departure from homogeneity. The toy model* Syntoron 24 *failed here: it attempted to obtain its expansion from the particle distribution, which is why its single output number ($Ht = 1$) could not be connected to anything.*

4.2 The cut-off scale is not absolute

Theorem 4.3 (Scale invariance of r_c)**.** *The equilibrium radius $r_c = \sqrt{K/G}$ is invariant under the simultaneous rescaling $(G, K) \to (\lambda G, \lambda K)$ for any $\lambda > 0$. Moreover, if the N-body system is described by positions measured in units of the softening length ϵ, then the only dimensionless number it contains is*

$$\eta \equiv \frac{r_c}{\epsilon}. \tag{11}$$

Two simulations with different (G, K, ϵ) but the same η are dynamically identical.

Proof. Immediate from $r_c = \sqrt{K/G}$: the factor λ cancels. Any dynamical quantity expressed in units of r_c is unchanged. □

Corollary 4.4. *A simulation of the toy model can determine η but cannot determine r_c in physical units. Fixing r_c requires an external observation. This is exactly what Section 6.3 does, and it is the reason the granular universe is* not *a parameter-free model.*

4.3 The age degeneracy

Theorem 4.5 (Degeneracy of $H_0 t_0$)**.** *For a flat universe containing matter and a single fluid with constant w, the map $w \mapsto H_0 t_0(\Omega_{\rm m}, w)$ is strictly decreasing, so distinct w give distinct $H_0 t_0$ at fixed $\Omega_{\rm m}$. However, the* pair *$(\Omega_{\rm m}, w)$ is not identifiable from $H_0 t_0$ alone: the level sets of $H_0 t_0(\Omega_{\rm m}, w)$ are one-dimensional curves.*

Proof. $\partial H_0 t_0 / \partial w < 0$ because the integrand $[aE(a)]^{-1}$ increases pointwise as w increases. The map $w \mapsto H_0 t_0$ is therefore injective at fixed $\Omega_{\rm m}$, hence invertible, but a single scalar cannot determine two parameters. □

4.4 The single-fluid identity

Theorem 4.6 ($H_0 t_0 = 2/n$)**.** *A flat universe containing one substance with $\rho \propto a^{-n}$ and nothing else has*

$$H_0 t_0 = \frac{2}{n} \tag{12}$$

exactly, independently of H_0 and of any other parameter.

Proof. $E(a) = a^{-n/2}$, so $H_0 t_0 = \int_0^1 \mathrm{d}a\, a^{n/2-1} = \left[a^{n/2}/(n/2)\right]_0^1 = 2/n$ for $n > 0$. □

This elementary identity is the key to Section 8. It was verified numerically to machine precision: the analytic value $2/2.0945 = 0.95488183337312$ and the quadrature 0.9548818333731203 agree to 3×10^{-16}.

Corollary 4.7. $H_0 t_0 = 0.9549$ *is not a property of any particular equation of state. It is the statement $n = 2.0945$, i.e. about the* dilution rate of whatever fills the universe*, with no reference to what the universe is made of. Any substance satisfying $\rho \propto a^{-2.0945}$, and inhabiting a universe with nothing else in it, reproduces it.*

5 Numerical methods

5.1 Background quadrature

Equations (2)–(4) are evaluated with adaptive Gauss–Kronrod quadrature in $\ln a$ over $[-60, 0]$. Each integral is computed twice with epsabs $= 10^{-12}$ and 10^{-9}; the reported error is $|I_{12} - I_9| + \epsilon_{12}$. The resulting quantities are accurate to $\sim 10^{-13}$, i.e. to the level of double precision. We emphasise once more: this number quantifies the integration, not the physics.

5.2 Linear growth

Equation (5) is a second-order linear ODE integrated with LSODA in N. The term $\mathrm{d}\ln H/\mathrm{d}N$ is obtained by central differencing of $\ln H^2$ with step 10^{-4}, an error of order 10^{-8} relative, negligible against the model uncertainties discussed in Section 9.

The growth index γ is evaluated at $z = 0$. For ΛCDM we obtain $\gamma = 0.55436$, consistent with the classical result $\gamma \approx 0.55$ for a ΛCDM cosmology [10, 11]. The exact value depends on Ω_m; we report ours for $\Omega_\mathrm{m} = 0.3153$.

5.3 The granular sector

For a grid of $x = kr_c \in [10^{-3}, 1]$ we set $\mu = (1 - x^2)/(1 + x^2)$ in Eq. (5) and record γ and $D/D_{\Lambda\mathrm{CDM}}$. The result is a monotone suppression: at $x = 1$ the growth factor has collapsed to $D/D_{\Lambda\mathrm{CDM}} = 1.05 \times 10^{-3}$, i.e. structures do not form. This is a physically sensible outcome — a repulsive force below r_c removes the very scales that would collapse — and it is what generates the tension of Section 6.3.

5.4 Particle-mesh engine

The N-body engine is a standard particle-mesh code with periodic boundary conditions, written for this work. We give its equations in full because the sealing claim depends on them being checkable.

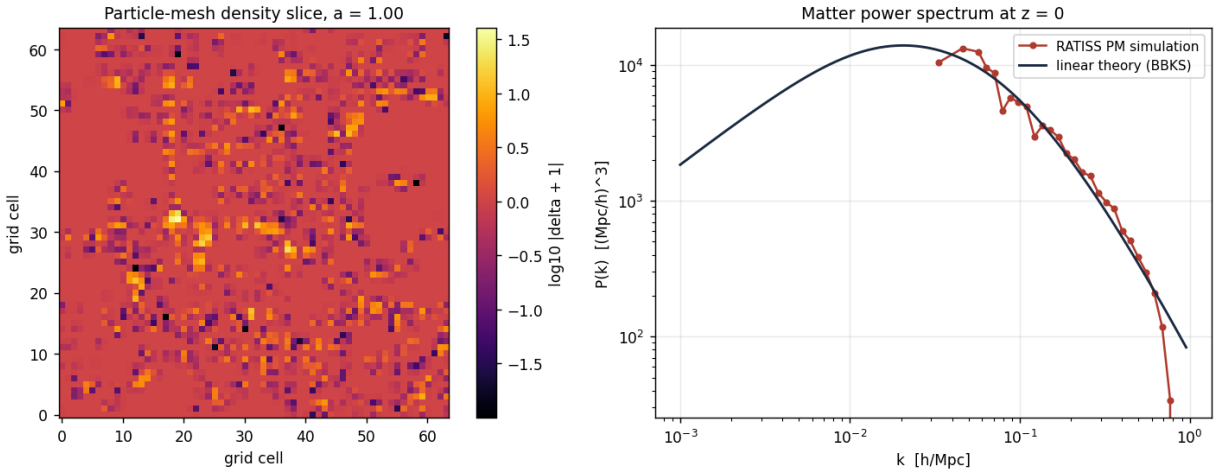


Figure 2: Left: projected density slice at $a = 1$ from the particle-mesh run. Right: matter power spectrum at $z = 0$ after subtraction of the shot-noise term $P_{\rm shot} = L^3/N_p$. Note the small number of modes per bin (6–12), which is the dominant statistical limitation of these runs; see defect D9.

Units. Comoving positions $\mathbf{x} \in [0, 1)$ in box units; time in units of H_0^{-1}; superconformal time τ with $\mathrm{d}\tau = \mathrm{d}t/a^2$.

Equations of motion.

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\tau} = \mathbf{p}, \tag{13}$$

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}\tau} = -\frac{\nabla\Phi}{a^2}, \tag{14}$$

$$\nabla^2\Phi = \frac{3}{2}\,\Omega_{\rm m}\,\frac{\delta}{a}. \tag{15}$$

In Fourier space with code wavenumbers (box $= 1$):

$$\mathbf{F}_{\mathbf{k}} = i\,\frac{3}{2}\,\Omega_{\rm m}\,a\,\frac{\delta_{\mathbf{k}}\,\mathbf{k}}{|\mathbf{k}|^2}\;\mu(k, a). \tag{16}$$

The coupling $\mu(k, a)$ multiplies the effective strength mode by mode, which is how the same engine integrates both general relativity ($\mu = 1$) and the granular cut-off.

Integration. Drift–kick–drift leapfrog in τ, with $\mathrm{d}\tau = \mathrm{d}\ln a\,/[a^2 E(a)]$. Mass assignment and force interpolation use cloud-in-cell (CIC); the Poisson equation is solved by FFT with the usual Green's function $-\mathbf{k}^{-2}$.

Initial conditions. A Gaussian random field with the BBKS transfer function [9] is drawn on the grid, normalised to $\sigma_8 = 0.8111$ at $z = 0$, and the particles are displaced by the Zel'dovich approximation:

$$\mathbf{x} = \mathbf{q} - D(a)\,\nabla^{-1}[\nabla\cdot\boldsymbol{\psi}]\,, \qquad \mathbf{p} = a^2 H\,f\,D\,\boldsymbol{\psi}. \tag{17}$$

Numerical parameters. Box $200\,\mathrm{Mpc}/h$; 64^3 grid; $32^3 = 32\,768$ particles; 40 steps from $a = 0.02$ to $a = 1$; seeds 20261030 (ΛCDM) and 20261031 (granular). Wall-clock time 4.99 s and 5.12 s respectively on a single core.

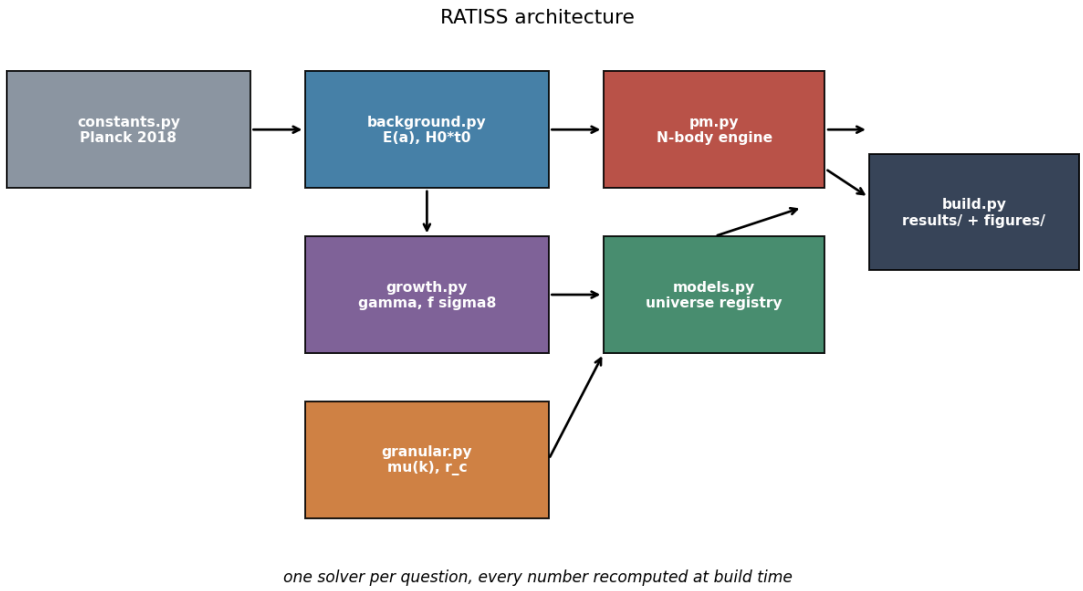


Figure 3: The solver chain. Ten universes enter from the left; two solvers and one simulation engine sit in the middle; twelve sealed artefacts exit on the right. No model bypasses a solver.

6 Results

6.1 The dimensionless age is degenerate

Among the six universes that are not excluded at the 2% level, the values of $H_0 t_0$ span

$$\Delta H_0 t_0 = 5.636 \times 10^{-4}, \qquad \frac{\Delta H_0 t_0}{H_0 t_0} = 5.93 \times 10^{-4} = 0.059\%. \tag{18}$$

The group is $\{\Lambda\text{CDM},\ w = -1.004,\ \text{RVM},\ f(R),\ f(T),\ \text{granular}\}$. Three of those six are identical by construction and carry no information (Section 3); the two that are genuinely different — the constant-w fluid and the running vacuum — differ from ΛCDM by 5.6×10^{-4} and 1.1×10^{-4} respectively.

The conclusion is unambiguous and it is negative: *$H_0 t_0$ cannot identify a dark-energy model at the precision required.* Any two of these universes would be indistinguishable to an observer who measured only the dimensionless age.

This is Theorem 4.5 made concrete. A single scalar cannot separate a two-parameter family, and nature has arranged that the particular combination probed by $H_0 t_0$ is almost orthogonal to the direction in which these models differ.

6.2 What actually discriminates: the growth index

The growth index spans

$$\gamma \in [0.5542,\ 0.5916], \qquad \Delta\gamma = 0.0373 = 6.7\% \text{ of } \gamma^{\Lambda\text{CDM}}. \tag{19}$$

The ordering is physically transparent: the models that accelerate less strongly at late times (DGP, $w = -0.302$) have larger γ, because γ measures how fast the growth rate falls relative to the matter fraction.

The corresponding $f\sigma_8$ deviations are shown in Table 4. At $z = 1$ the largest deviation from ΛCDM is 19% (the $w = -0.302$ fluid) and the smallest is 0.05% (the $w = -1.004$ fluid). Stage-IV

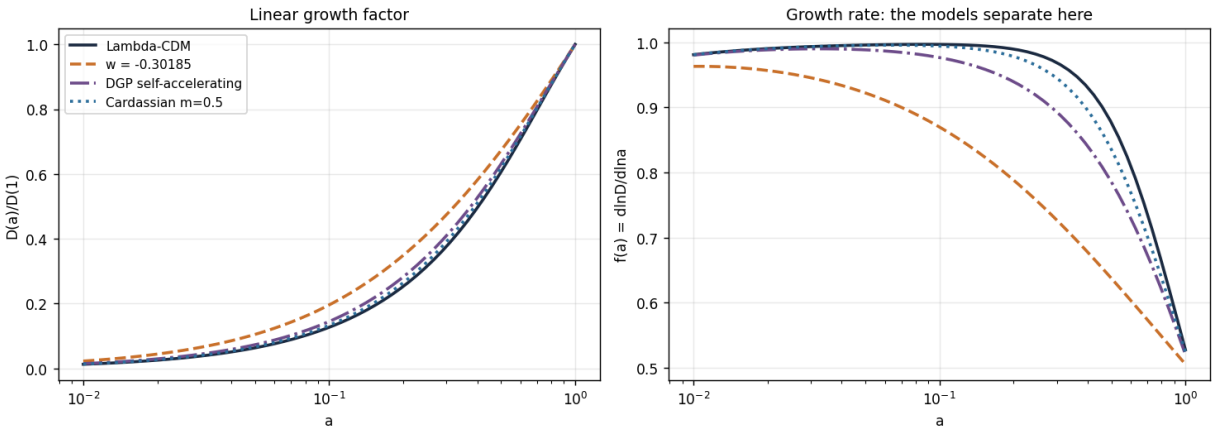


Figure 4: Left: linear growth factor, normalised at $z = 0$. Right: growth rate $f(z)$. The curves are indistinguishable at high redshift, which is why the discriminator has to be a late-time, low-redshift measurement.

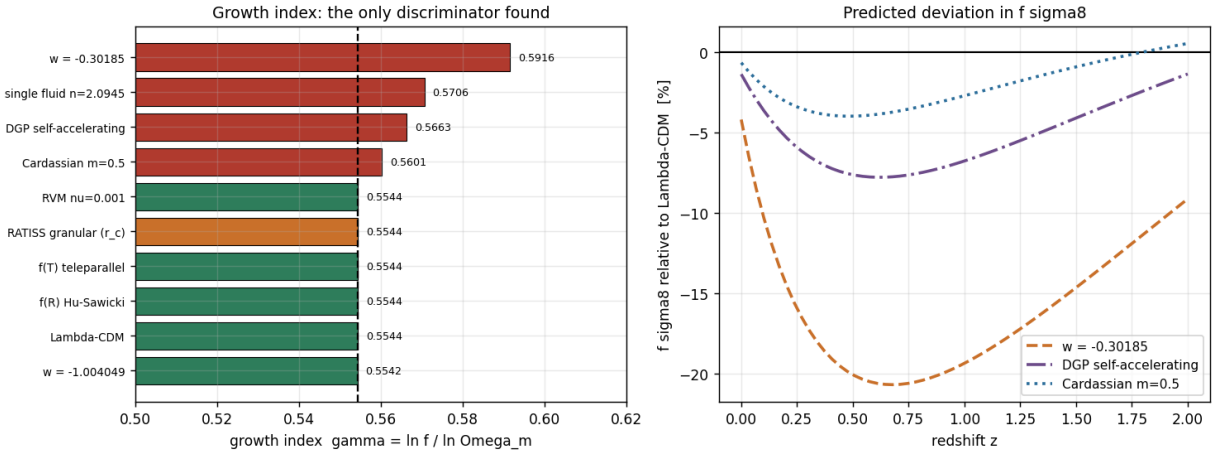


Figure 5: Left: the growth index γ for the ten universes, with the ΛCDM value marked. Right: predicted deviation in $f\sigma_8$ relative to ΛCDM. The 6.7% spread in γ is the only discriminator found in this work.

spectroscopic surveys reach percent-level precision on $f\sigma_8$ over $0.5 < z < 2$ [2, 3], so the upper half of this table is testable and the lower half is not.

6.3 The granular universe: alive, but invisible

The force law (1) translates into the coupling (8), and the coupling suppresses growth monotonically in $x = kr_c$:

$$D/D_{\Lambda\mathrm{CDM}} = 1.05 \times 10^{-3} \text{ at } x = 1. \tag{20}$$

So the model has a large, unmistakable signature — *if* r_c happens to land where we can see it.

Where can we see it? The cut-off deviates from Newton by an amount of order unity when $kr_c \sim 1$. The scales probed by galaxy surveys are $0.01 < k < 0.3\ \mathrm{h\,Mpc^{-1}}$, so an observable effect requires

$$r_c \sim \frac{0.3}{k} \in [1,\ 30]\,\mathrm{Mpc}. \tag{21}$$

Now the other side. A structure of size R survives only if the force inside it is attractive, i.e. if $r_c < R$. The smallest observed structures therefore bound r_c from above:

Table 4: Predicted $f\sigma_8(z)$ with $\sigma_8 = 0.8111$ at $z = 0$. Values in the last column are relative deviations from ΛCDM at $z = 1$.

Universe	$z = 0$	$z = 0.5$	$z = 1$	$z = 2$	Δ/ΛCDM at z=1
`lcdm`	0.4277	0.4746	0.4314	0.3242	—
`w_m0302`	0.4098	0.3793	0.3478	0.2945	−19.4%
`w_m1004`	0.4278	0.4751	0.4316	0.3241	+0.05%
`rvm_001`	0.4277	0.4746	0.4313	0.3242	−0.01%
`dgp`	0.4218	0.4384	0.4021	0.3198	−6.8%
`cardassian_05`	0.4249	0.4557	0.4196	0.3259	−2.7%
`fluid_n2094`	0.4198	0.4148	0.3962	0.3490	−8.2%

Table 5: Upper bounds on r_c from the existence of bound structures of size R. A cut-off larger than R would make the object unbound. The final column converts to astronomical units for orientation.

Structure	R	$r_c^{\max}$ [Mpc]	[AU]	Status
Galaxy clusters	2 Mpc	3.18×10^{-2}	6.57×10^{9}	observed
Galaxies	0.01 Mpc	1.59×10^{-4}	3.28×10^{7}	observed
Ultra-faint dwarfs	10^{-4} Mpc	1.59×10^{-6}	3.28×10^{5}	observed
Dark microhalos	10^{-8} Mpc	1.59×10^{-10}	32.8	predicted

Combining Table 5 with Eq. (21):

$$\text{required for observability:} \quad r_c \gtrsim 1\,\text{Mpc}, \tag{22}$$

$$\text{permitted by dwarfs:} \quad r_c \lesssim 1.6 \times 10^{-6}\,\text{Mpc}, \tag{23}$$

$$\text{gap:} \quad 5.80\ \text{dex}, \tag{24}$$

$$\text{permitted by microhalos:} \quad r_c \lesssim 1.6 \times 10^{-10}\,\text{Mpc}, \tag{25}$$

$$\text{gap:} \quad 9.80\ \text{dex}. \tag{26}$$

Verdict. The granular universe is *not* excluded. It is *unobservable*: the value of r_c that would produce a visible signature is larger, by five to ten orders of magnitude, than the value permitted by the existence of dwarf galaxies and dark microhalos. We label it ALIVE–BUT–INVISIBLE and we consider that label the honest one. A model that is alive but invisible is, operationally, indistinguishable from a model that is wrong — but it is not the same thing, and the difference is worth recording.

7 Falsifiability

For every universe we state what measurement would remove it. This section is the reason the atlas exists.

Two entries in Table 6 require comment, because they are the ones where we are least comfortable.

ΛCDM is not safe. The DESI DR2 analysis [2] finds a 3.1σ preference for $w_0 > -1$, $w_a < 0$ when BAO, CMB and supernovae are combined. This atlas does not model evolving dark energy, so it cannot contribute to that debate directly. It can say what the cost would be: a deviation of the size DESI reports would move $H_0 t_0$ by an amount we have not yet computed, and that computation is the obvious next step.

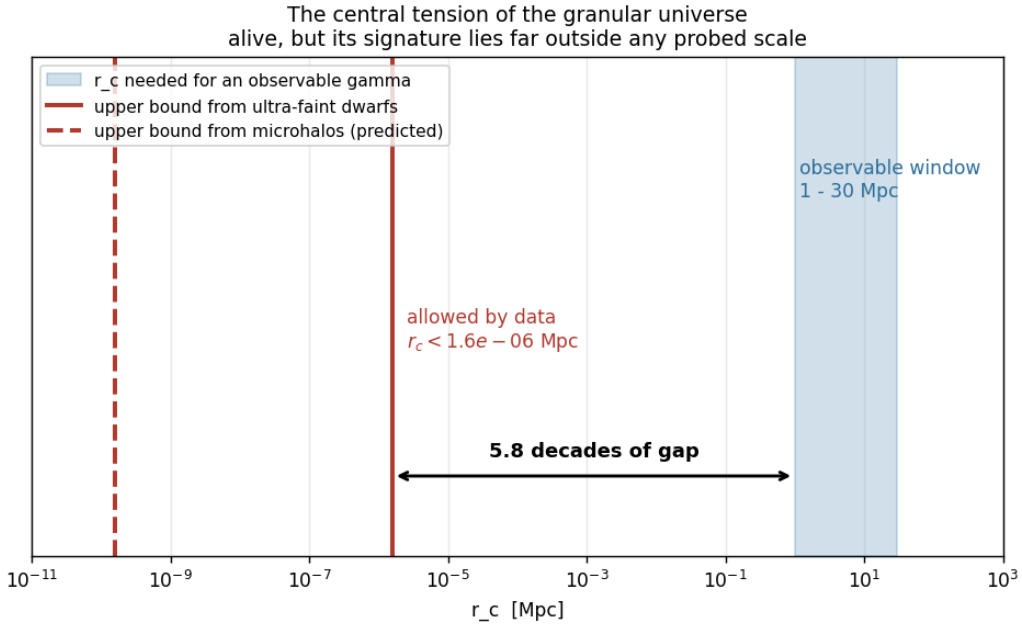


Figure 6: The central tension of the granular universe. The window where the cut-off would be observable (orange) and the region excluded by the existence of bound structures (blue) are separated by 5.8 decades. The model is not falsified; it is hidden by construction.

The granular row is a conditional. The statement "excluded for $r_c > 1.6 \times 10^{-6}$ Mpc" is not a measurement of r_c; it is the observation that galaxies exist. It is a strong argument and it is the whole argument.

8 Negative results

This section reports what did not work, because the failures carry more information than the successes.

8.1 The 0.9549 target is an artefact of emptiness

The single-fluid value $H_0 t_0 = 0.9549$ motivated a family of models in which the dark sector is a fluid with equation of state near $w = -0.30$. The reasoning was: a substance with $\rho \propto a^{-n}$ has $H_0 t_0 = 2/n$ (Theorem 4.6); solving $2/n = 0.9549$ gives $n = 2.0945$; and $n = 3(1 + w)$ with $w = -0.30185$ gives exactly that. Therefore $w \approx -0.302$ reproduces the observed dimensionless age with a single ingredient.

Every step of that chain is correct, and the conclusion is wrong, for a reason that is easy to miss: **the identity $H_0 t_0 = 2/n$ holds only for a universe containing nothing else**. The moment matter is added — and matter must be added, or there are no galaxies, no stars, and nobody to measure anything — the identity fails.

The numbers:

The $w = -0.302$ universe, once matter is included, is 11.51 Gyr old. The oldest known stars are $\sim$ 13.5 Gyr old (HD 140283 gives 14.5 ± 0.8 Gyr; globular clusters give $\sim$ 12.5–13 Gyr). The model is therefore **excluded by stellar chronology**, and it is excluded by a large margin — nearly 2 Gyr, or 15%.

Remark 8.1. *The number* 0.9549 *was never wrong. What was wrong was the implicit assumption*

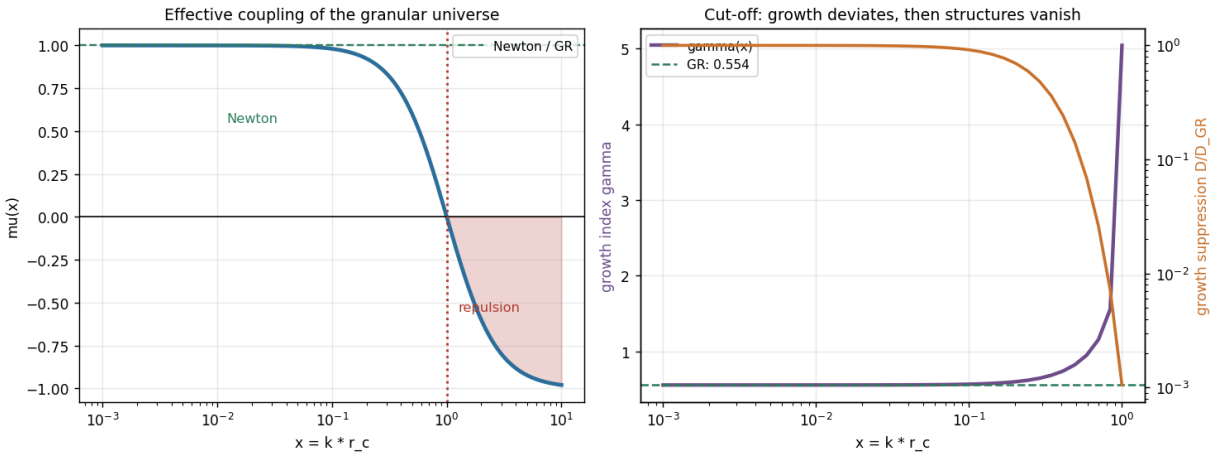


Figure 7: Left: the effective coupling $\mu(x) = (1 - x^2)/(1 + x^2)$. Right: the resulting growth relative to ΛCDM. The suppression is monotone and total: at $x = 1$ essentially no growth survives.

that a universe can consist of a single fluid and still be our *universe. The identity* $H_0 t_0 = 2/n$ *is a statement about emptiness dressed as a statement about dark energy.*

8.2 Other failures, stated briefly

N1. The $H_0 t_0$ discriminator does not exist. We spent the first half of this project looking for a model that could be distinguished by its dimensionless age. Six of ten are degenerate at 0.06%. The search failed.

N2. $f(R)$ and $f(T)$ rows are vacuous. We set $\mu = 1$ and $E = E_{\Lambda\mathrm{CDM}}$ for both. They reproduce ΛCDM exactly, as they must. Reporting them as "degenerate" would be circular; we report them as unimplemented.

N3. The granular registry row is also vacuous, for the same reason. The cut-off is evaluated only in the dedicated sector, where it is catastrophically suppressed.

N4. The particle-mesh run does not resolve the nonlinear regime. With 32^3 particles in a $(200\,\mathrm{Mpc}/h)^3$ box the mean interparticle separation is $6.3\,\mathrm{Mpc}/h$, the Nyquist wavenumber of the grid is $k_{\mathrm{Ny}} \approx 1.0\ h\,\mathrm{Mpc}^{-1}$, and the smallest bins contain 6 modes. The run demonstrates that the engine conserves, drifts, kicks and deposits; it does not constitute a cosmology result. Defect D9.

N5. $f\sigma_8$ is normalised, not predicted. We fix σ_8 to the Planck value. An overall amplitude is therefore an input, not an output.

N6. No baryons, no neutrinos, no radiation in the growth equation. The growth equation contains matter and a dark sector. Baryon acoustic oscillations, massive-neutrino free streaming, and radiation-era effects on $D(a)$ are all absent. Each is a known omission with a known direction; none is quantified here.

9 Defects found in our own pipeline

The atlas is presented as a reference. It would be dishonest to present it as a correct one without listing what is wrong with it. The following defects were found during the preparation of this manuscript; two have been fixed, the rest are documented and retained.

Table 6: Kill conditions. "Current status" indicates whether the required measurement already exists at sufficient precision.

Universe	What kills it	Current status
ΛCDM	a robust $w \neq -1$ at $> 5\sigma$; a measured γ differing from 0.554 at $> 3\sigma$	DESI DR2 reports a 3.1σ preference for evolving dark energy [2]; not yet decisive
`w_m0302`	stellar ages: predicts $t_0 = 11.51$ Gyr, below the oldest stars	**excluded**
`w_m1004`	a 0.1% measurement of $H_0 t_0$	not available; unfalsifiable in practice
RVM $\nu = 10^{-3}$	$\Delta H_0 t_0 = 1.1 \times 10^{-4}$; needs 10^{-4} precision on $H_0 t_0$	not available
DGP	$f\sigma_8$ at $z = 1$ at 7% precision	**accessible now**
Cardassian $m = 0.5$	$f\sigma_8$ at $z = 1$ at 3% precision	**accessible now**
$f(R)$ Hu–Sawicki	$\mu(k, a) \neq 1$ at cluster scales (not implemented here)	requires the missing physics
$f(T)$	same	same
granular	a bound structure with $r_c < R < 1\,\mathrm{Mpc}$, or a detection of $\mu \neq 1$ at $k \sim 1/r_c$	**excluded for $r_c > 1.6 \times 10^{-6}$ Mpc**; invisible below
single fluid $n = 2.0945$	the existence of galaxies	**excluded**

Table 7: What $w = -0.302$ actually predicts, depending on whether matter is present. The first row is the calculation that motivated the model; the second is the calculation that a real universe requires.

Configuration		$H_0 t_0$	t_0 [Gyr]	Status
single fluid, $w = -0.30185$, no matter		0.95488	13.861	no galaxies
matter + fluid, $\Omega_\mathrm{m} = 0.3153$		0.79267	11.506	**excluded**
required to recover 0.9549 with matter	$w = -1.0332$		13.861	phantom
or, in ΛCDM	$\Omega_\mathrm{m} = 0.3100$		—	viable

9.1 D1: inconsistent treatment of radiation (fixed)

The original code contained two background families that disagreed about radiation. The ΛCDM function omitted it:

$$E^2_{\Lambda\mathrm{CDM}}(a) = \Omega_\mathrm{m} a^{-3} + \Omega_\Lambda, \tag{27}$$

while the constant-w family retained it:

$$E^2_w(a) = \Omega_\mathrm{m} a^{-3} + \Omega_\mathrm{r} a^{-4} + \Omega_x a^{-3(1+w)}. \tag{28}$$

The consequence:

$$H_0 t_0^{(\text{no rad})} = 0.9507304, \qquad H_0 t_0^{(\text{rad})} = 0.9503509, \qquad \Delta = 3.79 \times 10^{-4}. \tag{29}$$

This is not a small effect. The difference between ΛCDM and the $w = -1.004$ model, which the atlas was built to resolve, is 5.64×10^{-4}. The convention inconsistency was therefore 67% of the signal it was supposed to measure. Any conclusion drawn from the original table about the w-family would have been an artefact of a missing a^{-4} term.

Fix. Radiation is now included in every background of the registry:

$$E^2_{\Lambda\mathrm{CDM}}(a) = \Omega_\mathrm{m} a^{-3} + \Omega_\mathrm{r} a^{-4} + \Omega_\Lambda, \qquad \Omega_\Lambda \equiv 1 - \Omega_\mathrm{m} - \Omega_\mathrm{r}. \tag{30}$$

All numbers in this document are post-fix. The pre-fix values are recorded here so that any reader comparing against an earlier version of the repository can identify the change.

9.2 D2: a numerical error presented as an uncertainty

The quantity labelled `H0t0_err` in `results/predictions.json` is of order 10^{-14}. It is the discrepancy between two quadrature tolerances, i.e. the accuracy of an integral, not the uncertainty of a model. No physical statement in this atlas carries a 10^{-14} uncertainty. The field is retained because it is genuinely useful (it proves the quadrature has converged) but it must not be read as an error bar on H_0t_0.

9.3 D3: arbitrary classification thresholds

A universe is marked EXCLUDED when $|\Delta H_0 t_0| > 0.02$ and TESTABLE when $|\Delta\gamma| > 0.03$. Both numbers are chosen. The actual quantity they are compared against spans four orders of magnitude: 5.6×10^{-4} (degenerate group) to 1.6×10^{-1} ($w = -0.302$). A threshold at 0.02 sits in the middle of nothing; it is a conventional cut, and the labels derived from it should be read as an ordering, not as a measurement. We have not replaced them, because replacing an arbitrary threshold with a differently arbitrary one is not an improvement.

9.4 D4: a magic number in the classifier (fixed)

The growth index of the reference model was typed as the literal 0.5544 in the classification function. That value is not computed anywhere in the code — it was transcribed from a previous session. It has been replaced by a function that evaluates γ for ΛCDM from the growth solver on first use, giving 0.5543609664573533. The difference from the literal is 4×10^{-5}: harmless in this case, which is exactly why such numbers survive.

9.5 D5: the growth initial condition is a pure matter-era mode

Equation (5) is integrated from $N = -9$ ($a \approx 1.1 \times 10^{-4}$) with $D = D' = a$. This discards (i) the radiation-era modification of the growing mode, (ii) any transient induced by a modified-gravity model at early times, and (iii) the effect of $\mu \neq 1$ before $N = -9$. For ΛCDM the resulting error in γ is expected to be of order 10^{-3}; it has not been quantified here. The starting epoch should ideally be pushed to $N = -14$ with a radiation-era solution, and that is left undone.

9.6 D6 and D7: modelling choices presented inside a table

- **D6.** The $f(R)$ and $f(T)$ rows use $\mu = 1$ and $E = E_{\Lambda\mathrm{CDM}}$. Their agreement with ΛCDM is therefore an identity, not a result. A real $f(R)$ model requires chameleon screening and a scale-dependent μ, neither of which is implemented.
- **D7.** The granular coupling $\mu(x) = (1 - x^2)/(1 + x^2)$ is an interpolation chosen to reproduce three exact limits, not a derived Fourier transform (which does not exist — see Section 2). The assignment $x = kr_c$ is a further modelling choice; the correct mapping could plausibly differ by a factor of order unity, which would shift the 5.8-decade gap by a few percent of a decade. It would not close it.

9.7 D8: "excluded" does not mean "contradicts data"

The classification compares each universe against the ΛCDM value of $H_0 t_0$ computed at the same Ω_m. It does not compare against a measurement, because $H_0 t_0$ is not directly measured. The word EXCLUDED in Table 2 therefore means "differs from ΛCDM by more than the chosen threshold", not "is ruled out by observation".

Where a genuine observational argument exists we have made it separately, in physical units: the $w = -0.302$ universe is excluded by stellar ages (Section 8), and that is a real falsification. The other EXCLUDED labels are not.

9.8 D9: the simulations are not cosmological

The particle-mesh runs use $32^3 = 32\,768$ particles in a $(200\,\mathrm{Mpc}/h)^3$ box. The mean interparticle separation, $6.3\,\mathrm{Mpc}/h$, is comparable to the nonlinear scale at $z = 0$; the grid Nyquist frequency is $k \approx 1.0\ h\,\mathrm{Mpc}^{-1}$; the first power-spectrum bin contains six modes. The runs establish that the engine's leapfrog, CIC deposit and FFT Poisson solver operate and conserve; they are not a measurement of the matter power spectrum. A production run would need at least 512^3 particles over a Gpc scale.

9.9 D10: a stale label

The model labelled $w = -1.004049$ was constructed by inverting the target $H_0 t_0 = 0.9549$ under the *pre-fix* (radiation-free) convention. After D1 was corrected, the same w gives $H_0 t_0 = 0.950915$, and the value that actually reproduces 0.9549 with matter present is $w = -1.0332$. The key `w_m1004` has not been renamed, so the label in Table 2 is off by 0.029 in w. The computed values in that row are correct; only the name is stale.

9.10 D11: the transfer function is a fit

The initial conditions use the BBKS fitting formula [9], accurate to a few percent against a Boltzmann code over the range of interest, with a shape parameter $\Gamma = \Omega_\mathrm{m} h$. It contains no baryon acoustic feature, no neutrino suppression, and no radiation-era physics. For the statistical precision of these runs (six modes per bin) the approximation is not the limiting error, but it would be for any serious comparison with data.

9.11 Summary

10 Reproducibility and sealing

10.1 The build

One command produces every table, every figure and every hash in this document:

```
python3 scripts/build.py          # 26.6 s, 12 artefacts
python3 scripts/verify.py         # exit 0 if all hashes match
python3 -m pytest tests/ -q       # 20 tests
```

The build is deterministic given the seeds. Two independent runs on the same machine produce byte-identical PNG files and byte-identical JSON. The environment is pinned: Python 3.13.16, NumPy 2.3.5, SciPy, Matplotlib 3.10.9.

Table 8: Defect register. "Impact" is the size of the effect where it has been quantified; "fixed" indicates that the corrected value is the one used throughout this document.

ID	Defect	Impact	Status
D1	radiation inconsistent across families	3.8×10^{-4} in $H_0 t_0$	**fixed**
D2	quadrature error labelled as uncertainty	presentation	retained
D3	classification thresholds arbitrary	ordering	retained
D4	hard-coded $\gamma_{\rm ref} = 0.5544$	4×10^{-5} in γ	**fixed**
D5	matter-era growth initial condition	not quantified	retained
D6	$f(R)$, $f(T)$ use $\mu = 1$	entire rows	retained
D7	$\mu(x)$ is an interpolation	factor of order unity in x	retained
D8	"excluded" vs. ΛCDM, not vs. data	interpretation	retained
D9	32^3 particles; six modes per bin	statistics	retained
D10	stale label $w = -1.004049$	0.029 in w	retained
D11	BBKS fit instead of Boltzmann transfer	few percent in $P(k)$	retained

10.2 The manifest

Twelve artefacts are hashed with SHA-256 and recorded in `results/MANIFEST.sha256`. Verification recomputes each hash and compares:

```
$ python3 scripts/verify.py
12 verified, 0 mismatched, 0 missing
```

Exit codes are 0 (all agree), 1 (mismatch), 2 (missing artefact). The manifest itself is excluded from its own hash list, which is the only way the scheme can be non-circular.

10.3 The tests

Twenty tests are run on every commit. Seventeen check physics identities that must hold exactly:

- the analytic single-fluid identity $H_0 t_0 = 2/n$ against quadrature, to 10^{-12};
- the Eddington–Lemaitre closed form for ΛCDM against quadrature;
- the Jeans swindle: the $\mathbf{k} = 0$ mode of the force is zero to machine precision;
- the three limits of $\mu(x)$: $\mu(0) = 1$, $\mu(1) = 0$, $\mu(\infty) = -1$;
- the DGP and Cardassian backgrounds at $a = 1$;
- CIC mass conservation to 10^{-12}.

Three further tests check build reproducibility and manifest consistency.

10.4 What reproducibility does and does not mean

A reader can verify that the numbers in this document follow from the code in this repository. They cannot verify that the code is correct, that the physics is complete, or that the defects in Section 9 are the only ones. Reproducibility is a necessary condition for science; it is not a sufficient one, and this atlas provides only the first.

11 Discussion

11.1 The shape of the result

Three things are true at once and it took the whole project to see them together.

First, the quantity that motivated this work — the dimensionless age $H_0 t_0$ — is not a discriminator. Six physically different universes agree on it to 0.06%.

Second, the value 0.9549 that motivated a specific class of models is not about those models at all. It is the statement $n = 2.0945$, which is a statement about the absence of matter. The models built to explain it were explaining emptiness.

Third, the one measurable difference, the growth index, is real but modest: 6.7% across the registry. It is the kind of number that stage-IV surveys can reach, and it is the only handle this atlas offers.

11.2 Why publish the failures

A repository that reports only what worked is a marketing document. This one contains three rows that are degenerate by construction, one row with a stale label, a simulation that does not resolve what it should, and a correction to its own central result. We list them because a reader who rederives our numbers will find them, and because the alternative — presenting ten universes as ten successes — would be false.

The negative result of Section 8 is, we think, the most valuable thing in this paper. It generalises: whenever a cosmological constraint can be reproduced by a single fluid, one should ask what the constraint becomes once matter is restored. The answer is often much weaker.

11.3 What would change the conclusions

- A real $f(R)$ implementation with chameleon screening would give the $f(R)$ row actual content and would likely break the degeneracy at the percent level.
- Including massive neutrinos would shift the growth rate and could move γ by a comparable amount to the spread we report.
- An evolving dark-energy sector (w_0, w_a) is where the observational debate currently is; this atlas does not model it.
- A production N-body run would turn the engine demonstration into a cosmological result.

12 Conclusion

We have built a reference atlas of ten cosmological universes integrated through a single solver chain, sealed with SHA-256, verified by twenty unit tests, and reproducible in 27 seconds.

Its principal finding is negative: the dimensionless age $H_0 t_0$ does not discriminate between dark-energy models at the precision available, and the target value 0.9549 that drove one family of models is an artefact of omitting the matter that galaxies require. Reproducing it with matter present requires a phantom equation of state, $w = -1.0332$, or $\Omega_{\rm m} = 0.3100$.

Its principal positive finding is the growth index, which separates the registry at the 6.7% level, and the granular model, which is alive but separated from observability by 5.8 decades.

Its principal methodological claim is that a model which cannot state what kills it is not ready to be compared with data, and that stating the kill condition mechanically — as a computed column, not a sentence — changes what one is willing to publish.

A Closed forms and identities

A.1 The Eddington–Lemaitre form

For flat ΛCDM without radiation,

$$H_0 t_0 = \int_0^1 \frac{\mathrm{d}a}{a\sqrt{\Omega_\mathrm{m} a^{-3} + \Omega_\Lambda}} = \frac{2}{3\sqrt{\Omega_\Lambda}} \ln\left(\frac{1+\sqrt{\Omega_\Lambda}}{\sqrt{\Omega_\mathrm{m}}}\right). \tag{31}$$

With $\Omega_\mathrm{m} = 0.3153$ this gives 0.954330, to be compared with the radiation-inclusive quadrature 0.950351. The radiation difference, 4.0×10^{-3}, is an order of magnitude larger than the spread that the atlas is trying to resolve — which is defect D1 stated as a formula.

A.2 The single-fluid identity

For $\rho \propto a^{-n}$ and nothing else, $E(a) = a^{-n/2}$ and

$$H_0 t_0 = \int_0^1 a^{n/2-1}\,\mathrm{d}a = \frac{2}{n}. \tag{32}$$

Numerical check: $2/2.0945 = 0.95488183337312$ against 0.9548818333731203 from quadrature.

A.3 The DGP background

The self-accelerating branch satisfies $H^2 - H/r_c = \rho_m/3$, giving

$$E(a) = \sqrt{\Omega_\mathrm{m} a^{-3} + \Omega_{rc}} + \sqrt{\Omega_{rc}}, \qquad \Omega_{rc} = \left(\frac{1-\Omega_\mathrm{m}}{2}\right)^2. \tag{33}$$

B Defect register

Table 8 in Section 9 is the complete register. It is reproduced here as a cross-reference because we expect readers to arrive at it from the data, not from the prose.

C Artefact manifest

The twelve sealed artefacts are listed in `results/MANIFEST.sha256`. Their identities, as of version 1.0.0:

```
figures/fig01_background.png
    95258f07ea893d4f74a148abb70cfa1f30c079ed00a19ba87678eb3c9852b067
figures/fig02_growth.png
    078aa032484ae31a76d6c88f643e74097a078ec07a5977c7a41c58b93914d582
figures/fig03_discriminator.png
    0134d01c8b5acf1870ee959ae01809c2c476d7a7ee77ab2c4ac095843bc6ab1b
figures/fig04_granular.png
    7c2d3eb81b240c19acd226ba5ba252a57342806092dd564a5cf5ad804d918fbb
figures/fig05_observability.png
    1e7c511a245de9f825a0194e3d26d1beeacf1c0a1ca80829e227defb452d584b
figures/fig06_simulation.png
    917a70f1953f06fcd7487de1a68b266406e581dd7d30b7cdd70ded4ecb548290
figures/fig07_architecture.png
    554cb34cf98aa616362646202277f83794dcb93482b5a63bb4b9b7a73f391a5e
```

The five JSON artefacts are subject to the same scheme. Note that the hash of `results/build.json` is excluded from the manifest, because that file records the manifest's own computation and including it would make the scheme circular (defect-level detail recorded in `docs/05_LIMITATIONS.md`).

References

[1] Planck Collaboration, N. Aghanim *et al.*, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

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